

# Lesson 6-5 · Quick

Properties of Logarithms + DOK-3 Power Property Proof &  
Richter Generalization

Klimesara-adapted · Student-centered · No Blooket

DOK 1–2

7 min

*bridge to log properties launch*

## 3 items: error analysis + expand + condense

1. **Error Analysis** A student approximates  $\log_2 9$  by entering  $\frac{\ln 2}{\ln 9}$  on her calculator. Why is this wrong?
2. **Expand:**  $\log_6 \frac{49}{5}$
3. **Condense:**  $5 \ln s + 6 \ln t$

# Launch — Example 3: Condense into a Single Log

Algebra 2 · Lesson 6-5 · Quick

DOK 1–2

5 min

teacher models

**Rules Card (keep visible all period)** **PRODUCT:**  $\log_b(MN) = \log_b M + \log_b N$

**QUOTIENT:**  $\log_b(M/N) = \log_b M - \log_b N$

**POWER:**  $\log_b(M^n) = n \cdot \log_b M$

**CHANGE OF BASE:**  $\log_b x = (\log x)/(\log b)$

**Condensing order: POWER first → then PRODUCT / QUOTIENT**

**Teacher models:**  $4 \log_4 m + 3 \log_4 n - \log_4 p$   
Step 1 (Power):  $\log_4(m^4) + \log_4(n^3) - \log_4 p$

Step 2 (Product then Quotient):  $\log_4\left(\frac{m^4 n^3}{p}\right)$

DOK 2–3

28 min

student work

6 items in order. Plan ~12 min for #13 + #40. Wed-F: skip #3.

**Practice #19 · Condense**  $\log_5 6 + \frac{1}{2} \log_5 y$

**Practice #21 · Condense**  $\frac{1}{3} \ln 27 - 3 \ln(2y)$

**Practice #34 · Solve (change of base)**  $7^x = 100$

**Practice #3 · Error analysis (skip Wed-F)**  $\log_4(c^2 d^5) = 2 \cdot \log_4 c \cdot 5 \cdot \log_4 d$

DOK 2

5 min

*DOK-3 reveal + Topic 6 bridge*

**Sentence frame · Power Property Proof (#13)** “Let  $x = \log_b M$ . Then  $b^x = M$ . Raise both sides to  $n$ :  $(b^x)^n = M^n$ , so  $b^{(nx)} = M^n$ . By the definition of log,  $\log_b(M^n) = nx = n \cdot \log_b M$ .”

**Sentence frame · Richter (#40 Part C)**  $R(150A_0) - R(A_0) = \log(150A_0/S) - \log(A_0/S) = \log 150 \approx 2.18$ .  
The larger earthquake is about **2.18 Richter units** greater.

**Topic 6 Assessment Bridge** #13 proof + #40 Part C = the exact DOK 3 pattern

DOK 1–2

5 min

*not graded*

**Complete on your exit ticket** A biologist models a population as  $P = 3 \log t + \log 4$ .

1. Write  $P$  as a single logarithm:  $P = \log(\underline{\hspace{2cm}})$
2. Solve for  $t$  when  $P = 2$ :  $t = \underline{\hspace{2cm}}$
3. One biggest thing I learned today:  $\underline{\hspace{4cm}}$